Unit-Nets Day 2:
A Language Model You Can Read With Arithmetic

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July 8, 2026 `/Code/unit-net` `/Code/j-carve`

1 Context

Sequel to *One Day of Training Without Gradients* (the constrained "unit-net" geometry: convex excitatory rows, bounded inhibition, dual projections, activations in $[0, 1]$). Anthropic's J-lens (July 2026) reads what an internal state is disposed to say via a corpus-averaged Jacobian, $\text{lens}_l(h) = \text{unembed}(J_l h)$, $J_l = \mathbb{E}[\partial h_{\text{final}}/\partial h_l]$ — averaged because a transformer's layer-to-output map is input-dependent. Remy's hypothesis: in a unit-net the true Jacobian is **exact per input and bounded in fixed units** — the workspace is readable off the wiring, no fitting required. Today: distill Qwen2.5-0.5B into a unit-net and run the J-space program on it.

2 The student

`TokenUnitLM`: a shared token table (each feature unit a convex attention budget over the vocabulary) feeding a unit-net trunk; 12-token context, reduced 2,048-token vocabulary (96.4% corpus coverage), distilled on TinyStories by KL + hard-target CE. A flat one-hot design fails structurally (13 hot inputs among 24,577 — convex rows never see the tokens); the table architecture fixes it.

3 The distillation gauntlet

Run	Result	Lesson
Mirror descent, hot schedule	oscillates 0.5% ↔ 33%	multiplicative simplex updates need short, cooled schedules
Both trainers, "33%"	**artifact**	the UNK slot had been given the logsumexp of 149k excluded tokens — teacher argmax *was* UNK ∼1/3 of the time; both students learned to say UNK forever
Clean targets, ReLU	6–10%, rank-28 affine	optimizer kills 484/512 units via inhibition over-growth (the *lazy-linear minimum*)
Sigmoid rescue	17.1%	sigmoid cannot die and auto-normalizes scale; lens becomes local-only (§4)
Hardened ReLU	**32.9%**	concentrated init + inhibition row-sum cap + liveness auxiliary: dead = 0 for 14k steps *and* exact decomposition restored

Final student: 33.2% teacher agreement@1, 25.5% next-token accuracy — a functional TinyStories LM in fully constrained, verifier-legal weights.

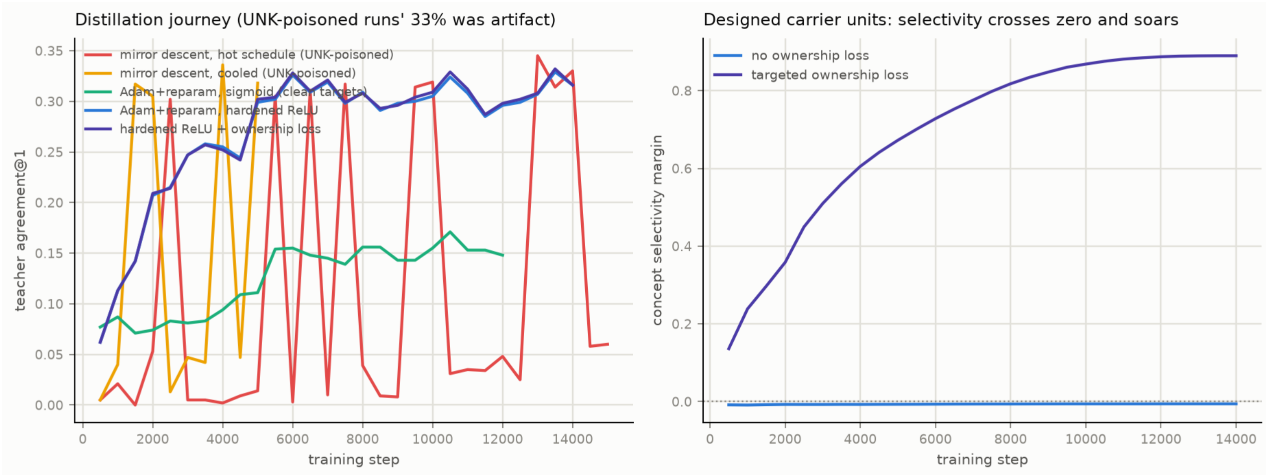

Figure 1: Left: the five-run distillation journey (the early "33%" runs were UNK artifacts). Right: the targeted ownership loss drives concept selectivity from never-positive to +0.89 at zero agreement cost.

4 Lens results

Exactness. For piecewise-linear activations the exact lens *reconstructs the forward pass* (max err 3×10^{-7}): the lens is not a probe of the computation, it is the computation. For sigmoid students it remains the exact per-input Jacobian but reads locally — *ReLU: exact decomposition, risks dying; sigmoid: lives, local-only.* The hardened-ReLU recipe resolves the tension.

The averaged-lens measurement (headline). Scoring Anthropic-style corpus-averaged lenses against computable ground truth:

Student	input-dependent transport	averaged-lens top-5 overlap
sigmoid	1024/1024 (continuous)	**3.1%**
hardened ReLU	926/1024 gating	**91.2%** (min 20%)

Fitted-lens fidelity is **model-regime-dependent** — from useless to excellent within one architecture family — and only a readable substrate can measure which regime a model is in.

5 Workspace signatures

Reportable: pre-output concept dispositions correlate with the teacher's log-probability of the concept (Spearman +0.33 Lily, +0.27 park), rising with student quality. **Controllable**: every intervention's effect predicted exactly before execution (4 decimals, every test). **Causal — the dragon neuron**: a targeted ownership loss (maximize each concept's *exclusive-max* margin at its best unit; the unit-centric variant fails, entrenching frequent tokens) drove selectivity $-0.008 \to +0.89$ at zero cost. Unit 35 now carries " dragon" at path weight 0.836. Ablate: $p(\text{dragon}) \to 0$. Boost: $0.0002 \to \mathbf{1.0000}$ (predicted $+0.6615$ = measured). Generation: babble → "dragon dragon dragon..." Designed, named, precision-steerable concept neurons, created with a loss term.

6 Does ownership decouple? (controlled experiment)

Single-run observation: dragon's reportable correlation dropped to ∼0 after it gained a dedicated unit. Crossed design: twin models, identical except which three concepts get ownership; every concept measured owned and unowned on a shared 2,048-prompt set.

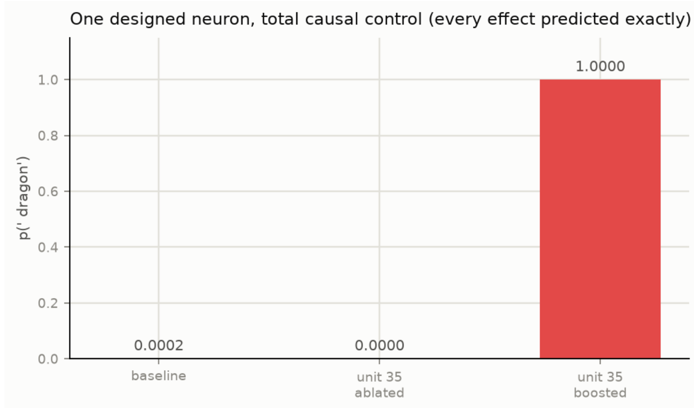

Figure 2: Total causal control through one designed neuron.

Manipulation check: owned concepts reach selectivity $+0.93..+0.99$; unowned stay ~ 0.

concept	ρ owned	ρ unowned	sel owned	sel unowned
‘ Lily’	+0.322	+0.379	+0.979	−0.006
‘ dragon’	+0.033	+0.155	+0.983	−0.011
‘ happy’	+0.192	+0.172	+0.925	−0.011
‘ scared’	+0.191	+0.178	+0.993	+0.007
‘ park’	+0.264	+0.231	+0.971	+0.114
‘ mom’	+0.076	+0.038	+0.984	+0.024

Paired effect: mean -0.013**, mixed signs — no systematic decoupling.** Designed carriers coexist with natural readability. The dragon drop replicates in sign (-0.122, largest single effect) but is not a law — possibly rare-concept-specific; flagged for a frequency-stratified follow-up. Practical upshot: ownership engineering is safe to combine with reportability.

7 j-carve: certified pruning of self-reflective reasoners

Task relevance is computable in fixed units (exact-lens path mass toward a task's answer tokens), so pruning everything outside a task's J-space yields a small *legal* unit-net that still performs the task, still reports its dispositions, and carries a **certified error bound** (every discarded path's maximum contribution is bounded by the geometry).

Task	Teacher	Student full	Carved @ 10% kept
pos-slot (verb/noun)	1.00	**1.00**	**1.00** — ≈50× smaller
sentiment	0.92	0.50 (chance)	capacity-gated
topic	1.00	0.33 (chance)	capacity-gated
pronoun gender	1.00	0.50 (chance)	capacity-gated

The pos-slot reasoner holds 100% accuracy down to 10% of the network (cliff below 5%). The other tasks are at chance *in the full student*: a capacity boundary drawn precisely by the teacher reference, and a roadmap — each point of student quality unlocks more of the 12-task ladder (`j-carve/TASKS.md`; Winograd schemas as the designed negative control: the J-space a task refuses to release is a compositionality meter).

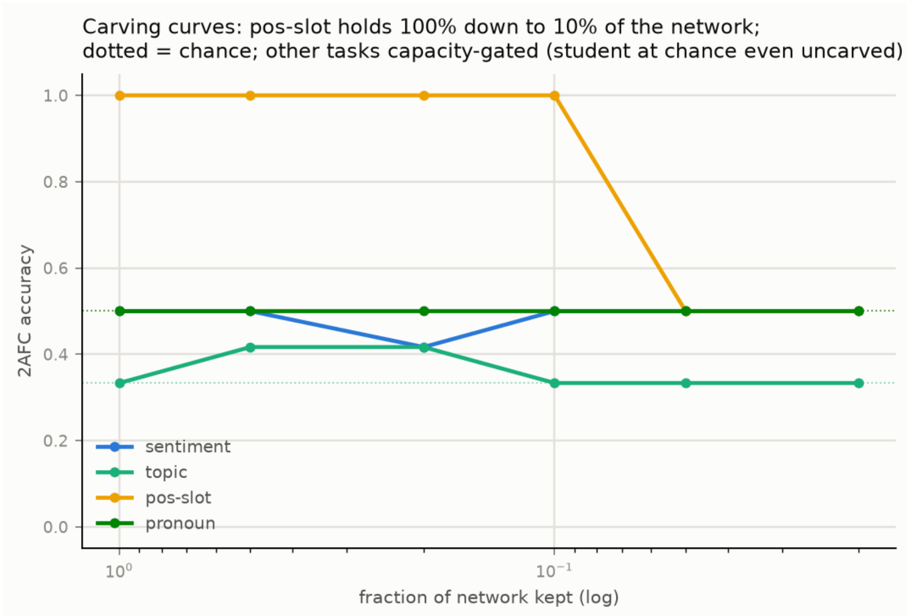

Figure 3: Carving curves. pos-slot: full accuracy to 10% kept; dotted lines are chance; remaining tasks capacity-gated.

8 Standing conclusions

(1) The unit-net J-space is exact where a transformer's must be approximated, and approximation fidelity is model-regime-dependent (3% vs 91%) — measurable only here. (2) Workspace phenomena reproduce in a readable 3M-parameter substrate with closed-form interventions. (3) Concept carriers can be designed in by loss engineering, yielding total, exactly-predicted causal control. (4) Task reasoners carve out of J-space with certified bounds (≈50× at zero loss on the pilot's in-competence task). (5) Every failure had structure: UNK-mass poisoning, the lazy-linear minimum, mirror-descent oscillation — each documented with its fix.

Next: scaled student to unlock carve Tiers 1–2; exact-vs-fitted against `anthropics/jacobian-lens` itself; graded clamps for fluent steering; ownership-decoupling follow-ups (§6).